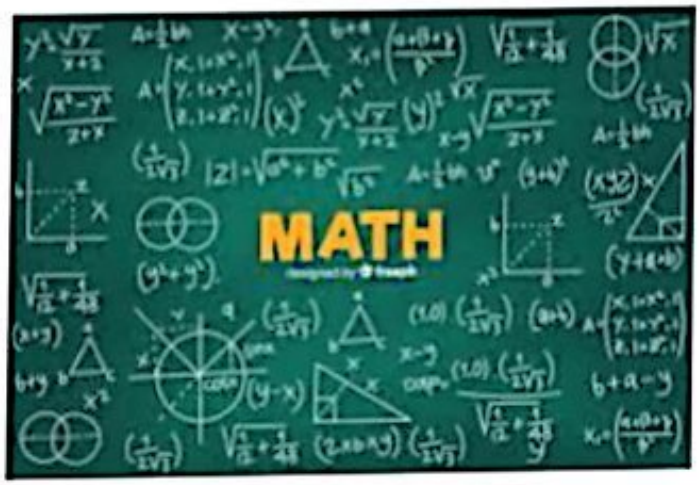


MATH 152 MID-SEMESTER EXAMINATION (2018/2019)

CALCULUS WITH ANALYSIS

ABSTRACT

This PDF file contains step-by-step solutions to the 2018/2019 MATH 152 mid-semester examination set by Dr Jeffery Ezeam.

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30

KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI

COLLEGE OF ENGINEERING

BSC AUTOMOBILE ENGINEERING, BSC CIVIL ENGINEERING,

BSC GEOMATIC ENGINEERING

SECOND SEMESTER MID-SEMESTER EXAMINATION (2018/2019)

FIRST YEAR (UNDERGRAD)

MATH 152—CALCULUS WITH ANALYSIS

MARCH 2019 TIME ALLOWED: 1 HR

Your index number: [redacted]

Programme of Study: BSc CIVIL ENGINEERING

Read the following **instructions carefully;**

1. Write your index number and programme of study CLEARLY at the top right of this page in the spaces provided.
2. Answer **ONLY ONE** Section! **CIRCLE** your answers with **INK** where required or **WRITE OUT** your answer in the space provided where required.
3. No communication without explicit permission from invigilation team.
4. Do not take any examination materials from the Examination Hall.

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(5) $\lim_{x\to 0} \frac{e^x-1}{x}$

Answer: 1

Evaluate the following limits.

(6) $\lim_{x\to 1} \frac{x^{99}-1}{x^3-1}$

A. 3 (B.) 33 C. 99 D. 1

(7) $\lim_{x\to 0} \frac{xe^{3x}}{\tan(\pi x)}$

A. $\frac{1}{\pi+e}$ (B.) $\frac{1}{\pi}$ C. $\frac{1}{e}$ D. $\frac{1}{\pi e}$

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(8) $\lim_{x\to 0} \frac{\sqrt{x+3}-\sqrt{3}}{x}$

A. $\sqrt{3}$ B. 0 C. $\frac{1}{2\sqrt{3}}$ D. The limit does not exist

(9) $\lim_{x\to\infty} \frac{x+2}{9x^2+1}$

Chosen answer for Q9 is A

A. 0 B. $\frac{1}{9}$ C. $\frac{2}{9}$ D. ∞

Use the function below to answer questions 10 and 11.

$$f(x) = \begin{cases} 7-4x & if\ x < 1 \\ x^2+2 & if\ x \geq 1 \end{cases}$$

(10) What is $\lim_{x\to -6} f(x)$?

A. 3 B. 16 C. 31 D. Does not exist

(11) What is $\lim_{x\to 1} f(x)$?

A. 3 B. 16 C. 31 D. Does not exist

Use the function below to answer questions 12 and 13.

$$g(x) = \begin{cases} 6x & if\ x \leq -4 \\ 1-9x & if\ x > 4 \end{cases}$$

(12) What is $\lim_{x\to 7} g(x)$?

A. 2 B. 7 C. -62 D. Does not exist

(13) What is $\lim_{x\to -4} g(x)$?

A. 2 B. 7 C. -62 D. Does not exist

(14) Which of the statements is neither true nor false?

A. If $\lim_{x\to a} \frac{f(x)-f(a)}{x-a}$ exists, then f is continuous at $x = a$.

B. If f is continuous at $x = a$, then it is differentiable there.

C. If $f'(a)$ exists, then $\lim_{x\to a} f(x)$ exists as well.

D. If $\lim_{x\to a} f(x)$ exists, $f'(a)$ may not exist.

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(15) Let $f(x) = \tan x$. Find $f''(x)$, the second derivative of $f(x)$.

A. $2\sec^2 x \tan x$ B. $2\sec x \tan^2 x$ C. $2\sec x \tan x$ D. $2\sec x^2 \tan^2 x$

SECTION II—Intermediate

Write out your answers for the first three questions.

(16) What is $\lim_{x\to 1} \frac{x^3-3x+2}{x^4-4x+3}$

Answer: _____

(17) What is $\lim_{x\to 1} \frac{\sin(x)-\sin(1)}{x-1}$

Answer: _____

(18) Find the derivative of $f(x) = x^x$

Answer: _____

(19) Find $g'(3)$ if $g(x) = x2^{h(x)}$ where $h(3) = -2$ and $h'(3) = 5$.

A. $\frac{1-15\ln 2}{4}$ B. $\frac{1+15\ln 2}{4}$ C. $\frac{1-15\ln 2}{8}$ D. $\frac{1+15\ln 2}{8}$

(20) Suppose that $f'(2) = 3$. What is $\lim_{x\to 2} \frac{f(x)-f(2)}{\sqrt{x}-\sqrt{2}}$.

A. $3\sqrt{2}$ B. $4\sqrt{2}$ C. $5\sqrt{2}$ D. $6\sqrt{2}$

(21) Find and simplify $f'(0)$ given

$$f(x) = (x + 1)(2x + 1)(3x + 1)(4x + 1)(5x + 1)$$

A. 13 B. 14 C. 15 D. 16

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(22) Find the two values of k for which the function $f(x) = e^{kx}$ satisfies the equation

$$f''(x) - f'(x) - f(x) = 0$$

A. $\frac{1\pm\sqrt{5}}{2}$ B. $\frac{1\pm\sqrt{3}}{2}$ C. $\frac{1\pm\sqrt{2}}{2}$ D. $\frac{1\pm1}{2}$

(23) Find the limit

$$\lim_{x\to 0} \frac{\sqrt{1+x}-1}{\sqrt[3]{1+x}-1}$$

A. $\frac{3}{2}$ B. $\frac{1+\sqrt{3}}{2}$ C. $\frac{1-\sqrt{2}}{2}$ D. 0

(24) Find the limit below

$$\lim_{x\to\infty} \frac{x-1}{x+1}$$

B. 1 B. -1 C. 0 D. 2

(25) What expression is the limit below?

$$\lim_{h\to 0} \frac{f(x+2h) - 2f(x+h) + f(x)}{h^2}$$

A. $f(x)$
B. $f'(x)$
C. $f''(x)$
D. None of the above

SECTION III—Advanced

Write out your answer for the first two questions.

(26) Find the following limit of the sequence

$$\lim_{x\to\infty} \frac{\sqrt{x}}{\sqrt{x+\sqrt{x+\sqrt{x}}}}$$

Answer: _____

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(27) Find the derivative of the function

$$f(x) = x^{\frac{2}{3}} \frac{1-x}{1+x^2} \sin^3(x) \cos^2(x)$$

Answer: _____

(28) Which of the following functions satisfies the equation below

$$x^2 f''(x) - 3x f'(x) + 4f(x) = 0$$

A. $x^2(1 + 10^{10} \ln x)$
B. $x^2(1 + 10^x \ln x)$
C. $x^2(1 + x^{10} \ln x)$
D. $x^2(1 + x^x \ln x)$

(29) What is the following limit.

$$\lim_{x \to 0} x^{\sin x}$$

A. 1
B. 2
C. 3
D. 4

(30) Evaluate the limit below

$$\lim_{n \to \infty} \left(\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots + \frac{1}{2^n} \right)$$

A. 1
B. 2
C. 4
D. 8

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(1) $\left(2 \times 7 \times \frac{1}{2}\right) 23.5$

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① $\lim_{x\to 1} \frac{\sin 5x}{x} = \frac{\sin 5(1)}{1}$

$= \sin 5$ (in rad)

$= -0.9589$

② $f(x) = \frac{\sin 5x}{x}$

Let $u(x) = \sin 5x$

$\Rightarrow u'(x) = 5\cos 5x$

Let $v(x) = x$

$\Rightarrow v'(x) = 1$

From quotient rule;

$f'(x) = \frac{vu' - uv'}{v^2}$

$= \frac{x(5\cos 5x) - 1(\sin 5x)}{x^2}$

$f'(x) = \frac{5x\cos 5x - \sin 5x}{x^2}$

③ $f(x) = e^{\ln e^{\ln x}}$

$f(x) = e^{\ln x}$

$f(x) = x$

(N.B. $e^{\ln x} = x$)

The antilog of a log of a function equals the function and vice versa

Hence, $f'(x) = 1$

④ $f(x) = \sqrt{\frac{x(x-1)}{x-2}} = \frac{[x(x-1)]^{1/2}}{(x-2)^{1/2}}$

Let $u = [x(x-1)]^{1/2}$

$\Rightarrow u' = \frac{1}{2}(2x-1)[x(x-1)]^{-1/2}$

$u' = \frac{2x-1}{2[x(x-1)]^{1/2}}$

Let $v = (x-2)^{1/2}$

$\Rightarrow v' = \frac{1}{2}(x-2)^{-1/2}$

$v' = \frac{1}{2(x-2)^{1/2}}$

$f'(x) = \frac{vu' - uv'}{v^2}$

$= \frac{(x-2)^{1/2}\left\{\frac{1}{2}(2x-1)[x(x-1)]^{-1/2}\right\} - [x(x-1)]^{1/2}\cdot\frac{1}{2(x-2)^{1/2}}}{[(x-2)^{1/2}]^2}$

$f'(x) = \frac{2x-1}{2(x-2)}\sqrt{\frac{x-2}{x(x-1)}} - \frac{1}{2}\sqrt{\frac{x(x-1)}{x-2}}$

$\Rightarrow f'(x) = \frac{1}{2}$

⑤ $\lim_{x\to 0} \frac{e^x - 1}{x}$

Apply l'hopital rule

$d/dx(e^x - 1) = e^x$

$d/dx(x) = 1$

$\Rightarrow \lim_{x\to 0} \frac{e^x - 1}{x} = \lim_{x\to 0} \frac{e^x}{1}$

$= e^0 = 1$

⑥ $\lim_{x\to 1} \frac{x^{99} - 1}{x^3 - 1}$

Apply l'hopital rule

$d/dx(x^{99} - 1) = 99x^{98}$

$d/dx(x^3 - 1) = 3x^2$

$\Rightarrow \lim_{x\to 1} \frac{x^{99} - 1}{x^3 - 1} = \lim_{x\to 1} \frac{99x^{98}}{3x^2}$

$= \lim_{x\to 1} 33x^{96}$

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$\Rightarrow \lim_{x\to 1} \frac{x^{99}-1}{x^3-1} = 33(1)^{96}$

$= 33(1)$

$= 33$

(7) $\lim_{x\to 0} \frac{xe^{3x}}{\tan(\pi x)}$

Apply l'hopital rule;

$d/dx(xe^{3x})$

$= x(3e^{3x}) + 1(e^{3x})$

$= e^{3x}(3x+1)$ — pdt. rule

$d/dx(\tan(\pi x)) = \pi \sec^2(\pi x)$

$\Rightarrow \lim_{x\to 0} \frac{xe^{3x}}{\tan(\pi x)} = \lim_{x\to 0} \frac{e^{3x}(3x+1)}{\pi \sec^2(\pi x)}$

$= \frac{e^{3(0)}[3(0)+1]}{\pi \sec^2[\pi(0)]}$

$= \frac{e^0(0+1)}{\pi \sec^2 0}$

$= \frac{1(1)}{\pi(1)}$

$\Rightarrow \lim_{x\to 0} \frac{xe^{3x}}{\tan(\pi x)} = \frac{1}{\pi}$

(8) $\lim_{x\to 0} \frac{\sqrt{x+3}-\sqrt{3}}{x}$

Apply l'hopital rule

$d/dx(\sqrt{x+3}-\sqrt{3})$

$= \frac{1}{2}(x+3)^{-1/2}$

$d/dx(x) = 1$

$\Rightarrow \lim_{x\to 0} \frac{\sqrt{x+3}-\sqrt{3}}{x}$

$= \lim_{x\to 0} \frac{1}{2\sqrt{x+3}}$

$= \frac{1}{2\sqrt{0+3}} = \frac{1}{2\sqrt{3}}$

(9) $\lim_{x\to\infty} \frac{x+2}{9x^2+1}$

$= \lim_{x\to\infty} \frac{\frac{x}{x^2}+\frac{2}{x^2}}{\frac{9x^2}{x^2}+\frac{1}{x^2}}$

$= \lim_{x\to\infty} \frac{\frac{1}{x}+\frac{2}{x^2}}{9+\frac{1}{x^2}}$

$= \frac{\frac{1}{\infty}+\frac{2}{\infty^2}}{9+\frac{1}{\infty^2}}$

$\lim_{x\to\infty} \frac{x+2}{9x^2+1} = \frac{0+0}{9+0}$

$= 0/9$

$= 0$

(10) Since $-6 < 1$, it is defined by $f(x) = 7-4x$

$f(-6) = 7-4(-6)$

$= 31$

(11) Since $1 \geq 1$, it is defined by $f(x) = x^2+2$

$f(1) = 1^2+2$

$f(1) = 3$

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12) since $7 > 4$, it is defined by $g(x) = 1 - 9x$

$g(7) = 1 - 9(7)$

$g(7) = -62$

13) $g(-4) = 6(-4) = -24$

$g(-4) = 1 - 9(-4) = 37$

for a limit to exist;

$\lim_{x \to -4^-} g(x) = \lim_{x \to -4^+} g(x)$

$= g(-4)$

Since $-24 \neq 37$,
the limit does not exist.

15. $f(x) = \tan(x)$

$f'(x) = \sec^2(x)$

$f''(x) = 2(\sec x \tan x)\sec^{2-1} x$

$f''(x) = 2\sec x \tan x \sec x$

$f''(x) = 2\sec^2 x \tan x$

16. $\lim_{x \to 1} \frac{x^3 - 3x + 2}{x^4 - 4x + 3}$

$= \lim_{x \to 1} \frac{(x-1)^2(x+2)}{(x-1)(x^3 + x^2 + x - 3)}$

$= \lim_{x \to 1} \frac{x^2 + x - 2}{x^3 + x^2 + x - 3}$

(Applying l'hopital)

$= \lim_{x \to 1} \frac{2x + 1}{3x^2 + 2x + 1}$

$= \frac{2(1) + 1}{3(1)^2 + 2(1) + 1}$

$= 0.5$

N.B. You can verify your answer in Q16 by putting x to a value say 0.99 which is close to 1.

17. $\lim_{x \to 1} \frac{\sin x - \sin 1}{x - 1}$

Apply l'hopital

$\Rightarrow \lim_{x \to 1} \cos(x)$

$= \cos 1$

18. Let $y = x^x$

$\ln y = \ln x^x$

$\Rightarrow \ln y = x \ln x$

Differentiate both sides;

$\frac{y'}{y} = x\left(\frac{1}{x}\right) + 1(\ln x)$

$\Rightarrow y' = y(\ln x + 1)$

but $y = x^x$

and $y' = f'(x)$

$\Rightarrow f'(x) = x^x(\ln x + 1)$

19. $g(x) = x\,2^{h(x)}$

Differentiate;

$\frac{g'(x)}{g(x)} = 2^{h(x)} + x h'(x) 2^{h(x)} \ln 2$

$g'(3) = g(3)\left[2^{h(3)} + x h'(3) 2^{h(3)} \ln 2\right]$

but $g(3) = 3 \cdot 2^{h(3)}$

$\Rightarrow g'(3) = 3 \cdot 2^{2h(3)}\left[1 + x h'(3) \ln 2\right]$

$= 3 \cdot 2^{2(-2)}\left[1 + 3(5) \ln 2\right]$

$\therefore g'(3) = \frac{1 + 15 \ln 2}{4}$

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20. $\lim_{x\to 2} \frac{f(x)-f(2)}{\sqrt{x}-\sqrt{2}}$

Apply l'hopital rule

$= \lim_{x\to 2} \frac{f'(x)}{\frac{1}{2}x^{-1/2}}$

$= \lim_{x\to 2} 2\sqrt{x}\, f'(x)$

$x = 2, f'(2) = 3$

$\Rightarrow \lim_{x\to 2} \frac{f(x)-f(2)}{\sqrt{x}-\sqrt{2}} = 2\sqrt{2}(3)$

$= 6\sqrt{2}$

21. Let $a(x) = x+1$
$b(x) = 2x+1$
$c(x) = 3x+1$
$d(x) = 4x+1$
$e(x) = 5x+1$

From the pdt. rule;

$f'(x) = a'bcde + ab'cde + abc'de + abcd'e + abcde'$

Shortcut

Since $a(x), b(x), c(x), d(x)$ and $e(x)$ are all linear function of the form $kx+1$, then, $f'(0)$ is given by;

$k_a + k_b + k_c + k_d + k_e$
$= 1+2+3+4+5$
$= 15$

22. $f''(x) - f'(x) - f(x) = 0$

$f(x) = e^{kx}$
$f'(x) = ke^{kx}$
$f''(x) = k^2e^{kx}$

$\Rightarrow k^2e^{kx} - ke^{kx} - e^{kx} = 0$
$e^{kx}(k^2 - k - 1) = 0$

but $e^{kx} \neq 0$ for all any real value of k

$\Rightarrow k^2 - k - 1 = 0$

Using the quadratic formula;

$k = \frac{1 \pm \sqrt{5}}{2}$

23. $\lim_{x\to 0} \frac{\sqrt{1+x}-1}{\sqrt[3]{1+x}-1}$

by applying l'hopital rule (differentiate both numerator and denominator)

$\lim_{x\to 0} \frac{3}{2}(1+x)^{1/6}$
$= \frac{3}{2}(1+0)^{1/6}$
$= \frac{3}{2}(1)$
$= \frac{3}{2}$

24. $\lim_{x\to\infty} \frac{x-1}{x+1} = \lim_{x\to\infty} \frac{\frac{x}{x} - \frac{1}{x}}{\frac{x}{x} + \frac{1}{x}}$

$= \frac{1 - \frac{1}{\infty}}{1 + \frac{1}{\infty}}$

$= \frac{1-0}{1+0} = 1$

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25. $\lim_{h\to 0} \frac{f(x+2h)-2f(x+h)+f(x)}{h^2}$

$= \lim_{h\to 0} \left(\frac{f(x+2h)-f(x+h)}{h^2} - \frac{f(x+h)-f(x)}{h^2} \right)$

$= \lim_{h\to 0} \frac{\frac{f(x+2h)-f(x+h)}{h} - \frac{f(x+h)-f(x)}{h}}{h}$

but $\frac{f(x+2h)-f(x+h)}{h} = f'(x+h)$

Also, $\frac{f(x+h)-f(x)}{h} = f'(x)$

$\Rightarrow \lim_{h\to 0} \frac{f'(x+h)-f'(x)}{h}$

$= [f'(x)]' = f''(x)$

26. $\lim_{x\to\infty} \frac{\sqrt{x}}{\sqrt{x+\sqrt{x+\sqrt{x}}}}$

Let $y = \left(\frac{x}{x+\sqrt{x+\sqrt{x}}} \right)^{1/2}$

$\ln y = \frac{1}{2} \ln \lim_{x\to\infty} \frac{x}{x+\sqrt{x+\sqrt{x}}}$

$\ln y = 1/2 \ln \lim_{x\to\infty} \left(\frac{1}{1+0} \right)$

$\ln y = 1/2 \ln \lim_{x\to\infty} 1$

$\ln y = 1/2 \ln 1$

$\ln y = 1/2 (0)$

$\ln y = 0$

$\Rightarrow y = e^0$

$y = 1$

28. Cauchy-Euler equations are of the form;

$ax^2 f''(x) + bx f'(x) + cf(x) = 0$

Auxilliary eqn;

$am(m-1) + bm + c = 0$

$m(m-1) - 3m + 4 = 0$

$m^2 - 4m + 4 = 0$

$(m-2)^2 = 0$

$m = 2$ or $m = 2$

General solution;

$y = k_1 x^m + k_2 x^m \ln x$

$\Rightarrow y = k_1 x^2 + k_2 x^2 \ln x$

k_1, k_2 are constants

$y = x^2 (k_1 + k_2 \ln x)$

Hence, option A $x^2 (1 + 10^{10} \ln x)$ satisfies the equation for $k_1 = 1, k_2 = 10^{10}$

29. $\lim_{x\to 0} x^{\sin x}$

Let $L = \lim_{x\to 0} x^{\sin x}$

$\ln(L) = \ln \lim_{x\to 0} x^{\sin x}$

$= \lim_{x\to 0} \sin x \ln x$

$= \lim_{x\to 0} \frac{\ln x}{\operatorname{cosec} x}$

This is indeterminate so we apply l'hopital rule

$= \lim_{x\to 0} \frac{\frac{1}{x}}{-\operatorname{cosec} x \cot x}$

$= -\lim_{x\to 0} \frac{\sin x \tan x}{x}$

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Apply l'hopital rule again

$$\ln L = -\lim_{x\to 0} \frac{\cos x \tan x + \sin x \sec^2 x}{1}$$

$$\Rightarrow \ln L = 0$$

$$L = e^0$$

Hence, $\lim_{x\to 0} x^{\sin x} = 1$

30. This can be considered as a G.P. with

$a = 1/2$

$r = \frac{1}{4} \div \frac{1}{2} = \frac{1}{2}$

$$S_\infty = \frac{a}{1-r}$$

$$= \frac{1/2}{1-1/2}$$

$$= 1$$

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